AI OPEN
DATA CENTER



Can the Industry Sustain AI?
The Water-Energy Nexus Debate



Brian Jabeck, MODERATOR *SVP, Business Development, DLB Associates & Affiliate Companies*

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Rob Coyle *Senior Director of Member Experience, Open Compute Project Foundation (OCP)*

Michael Obradovitch *Vice President Global Accounts, Nalco Water (Ecolab)*

Megan Schneider *Enterprise & Industry Transformation (Energy / Utilities), Accenture*

Panel Discussion



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Why Water, Why Now

- **Hotter summers, stressed grids** — communities feel it first: higher utility bills, shrinking freshwater reserves
- **Data centers are the visible face of AI** — public scrutiny lands on the building, not the cloud.
- **Water and energy are one system —**Water is visible. For cooling systems, water and energy are inextricably linked.
- **Water is the deciding factor** — siting, permitting, and public acceptance.



WRI Aqueduct Overall Water Risk — where demand meets scarcity

100×
on-site water range across cooling designs

~10×
PJM capacity price rise in two years

0–5
WRI basin risk tiers driving siting

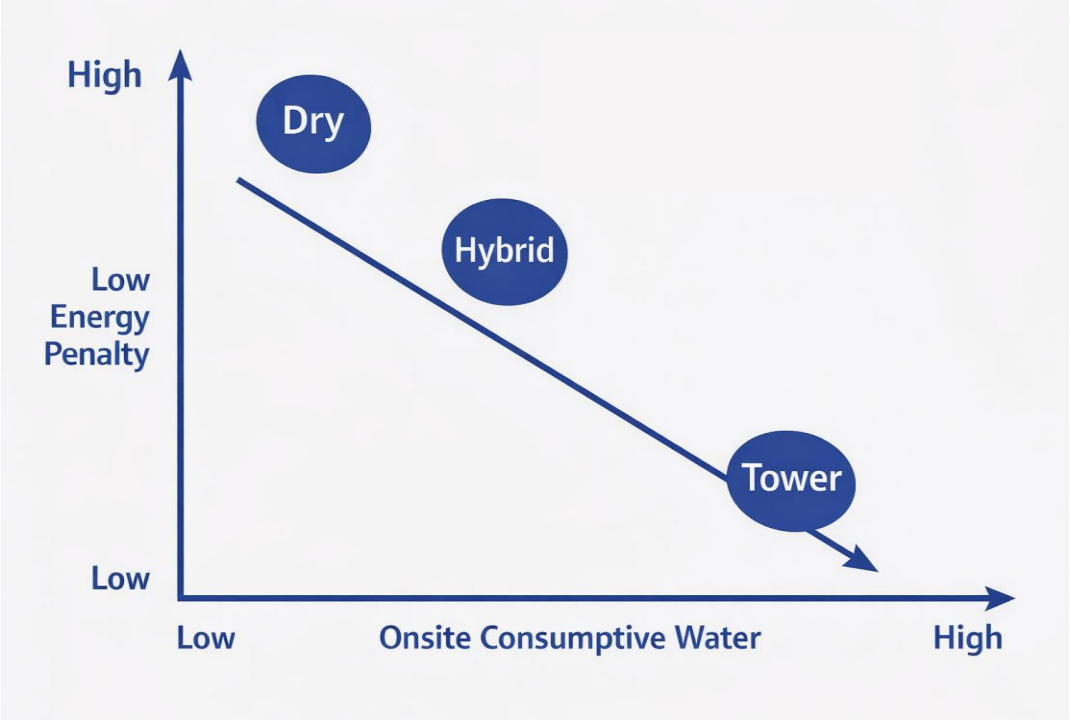
Sources: ASHRAE Journal series Parts 1–3 (Quirk et al., 2026); "AI and the Water-Energy Nexus"; WRI Aqueduct.

The Nexus Defined: Site Water vs. Source Water

- **No single "water number".** Cooling System design and power generation determines water usage.

Cooling System	Site Water — WUE (gal/MWh)	Source Water — (gal/MWh)	Total: Site + Source (gal/MWh)
Air-cooled	0	300–500	300–500
Adiabatic	0–105	300–500	300–605
Evaporative	450–740	300–500	750–1,240

- **To find to Water, follow the Source.** Data centers consume water twice: SITE (utility meter) + SOURCE (power generation).
- **"Going dry" isn't water free.** the water moves upstream and out of sight.
- **Engineer a Solution**. IT equipment with higher temperature tolerances and utilizing liquid cooling relaxes the water-energy trade.



Heat rejection is the variable — towers spend water; dry coolers spend energy, hybrids tune seasonally

Sources: ASHRAE Journal Part 1; CTI "Heat Rejection Physics" and "The Pareto Frontier."

Measuring What Matters: Accounting → Impact → Action

WUE

BASIC ACCOUNTING

On-site liters per IT kWh. Necessary but limited in scope: blind to the grid, the basin, and construction.

WUEsource

WHOLE SYSTEM

(EWIF × PUE) + WUE. Adds the power plant's water usage.

WUI

IMPACT

WUE × (stress factor + 1). Water is local. An Arizona gallon ≠ a Washington gallon.

WUC

ACTION

Water Use Credits: Conservation · Offset · Technology. Tradable, trackable.

Sources: ASHRAE Journal Parts 1–3; "Metrics Stack: WUE → WUI → WUC."

From Reporting to Outcomes: The WUC Flow Network

- **Basin-bounded cap-and-trade** — the watershed authority issues, transfers, and retires credits.
- **The power plant sits inside the boundary** — nobody wins by pushing water upstream
- **REC-grade guardrails** — registry, third-party verification, additionality, no double counting.

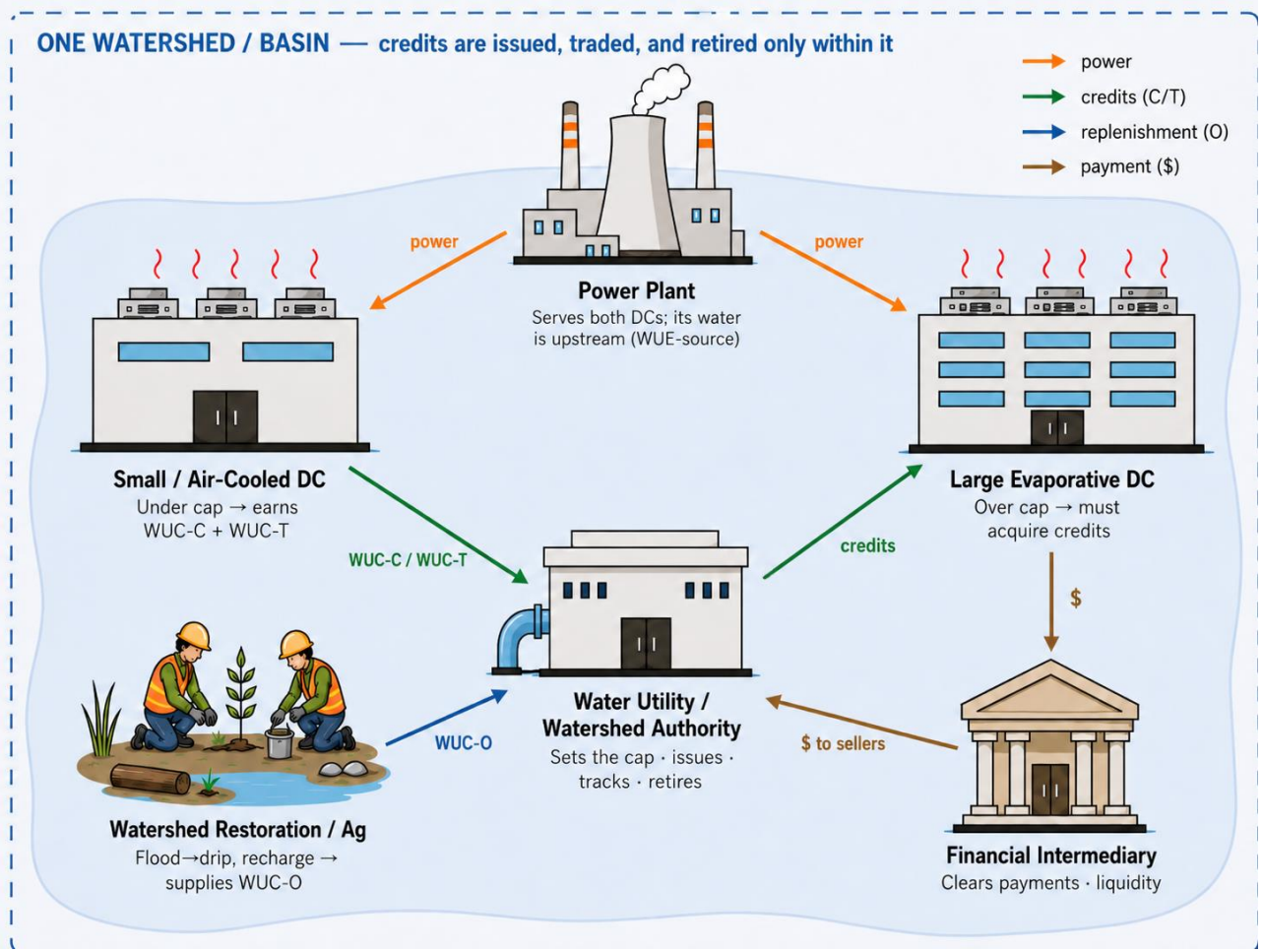


Figure 1, ASHRAE Journal Part 3;
"The WUC Flow Network: A Cap-and-Trade Structure for Water."

What's Still Unsolved: the Panel's Territory

Construction water is invisible

Construction and start-up water demand spikes hit communities. The industry is reactive. There is no accounting. No accountability and no plan.

Water Pricing Excludes Scarcity

Utility rates recover infrastructure, not local scarcity: San Diego (extreme stress) pays $9.36/CCF while Milwaukee (mid-high) pays $1.69.

Voluntary isn't verifiable

Every hyperscaler is “water positive by 2030” by its own definition. Self-defined metrics, no common registry, additionality unresolved.

Trust is the scarcest resource

Projects are won or lost at county hearings. Exaggerated numbers circulate on both sides, and one overstated claim costs an advocate the room.

Sources: ASHRAE Journal Part 3; DLB “Data Centers and Their Neighbors” white paper.

Open Discussion



Call to Action

- The industry needs an actionable framework and standardized playbook to reduce consumptive water while reducing energy waste (numerically lower WUE and lower PUE)
- We need a credible path to adopt WUC (Water Usage Credits) with both credits (avoidance) and offsets (replenishment) - modeled after RECs

How to get involved:

- OCP: A Call for Collaboration on AI Data Center Infrastructure Standards https://www.opencompute.org/about/a-call-for-collaboration-on-ai-data-center-infrastructure-standards
- iMasons Climate Accord: https://imasons.org/initiatives/climate-accord/
- ASHRAE: https://www.ashrae.org/technical-resources/ashrae-journal